SANJIVANI UNIVERSITY

SCHOOL OF ENGINEERING AND TECHNOLOGY

At: Kopargaon, Dist: Ahmednagar, Maharashtra, India — 423601

DEPARTMENT (Artificial Intelligence and Data Science)



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Telecom Customer Churn Prediction using Machine Learning

Project title	Telecom Customer Churn Prediction using Machine Learning
Team Name	CodeCrafter
Team Members	Vaishnavi Bhise(2124UDSF1006) Srushti Jamdar(2124UDSF1030) Pranjal Dhamane(2124UDSF1035)
Guide / Faculty	Rohit Pawar Sir
Academic year	2026-27
GitHub repository	https://github.com/VaishnaviBhise05/telecom-churn-prediction.git

Abstract

Customer churn, the loss of customers to competitors, is a major source of revenue loss for telecom companies. This project builds an end-to-end machine learning solution that predicts which customers are likely to leave, so that the company can run targeted retention campaigns. Using the IBM Telco Customer Churn dataset (7,043 customers, 21 attributes), we performed data cleaning, statistical analysis, exploratory data analysis, feature engineering, and compared Logistic Regression, Random Forest and Gradient Boosting models using stratified 5-fold cross-validation. The final tuned Gradient Boosting model, with a business-driven decision threshold of 0.32, identifies 75% of churners on unseen data (recall 0.75, F1 0.63, ROC-AUC about 0.85). A simulation shows a net benefit of about 39,580 currency units on a 1,409-customer test group. The project concludes with actionable recommendations and a Streamlit web app for live demonstration.

1. Problem Definition and Business Understanding

1.1 Problem statement

A telecom operator loses revenue every time a subscriber cancels the service. Acquiring a new customer costs several times more than retaining an existing one, yet the company cannot tell in advance which customers will leave. The problem is to predict, from a customer's profile, services and billing details, whether the customer will churn.

1.2 Objectives

1. Identify the key factors that drive customer churn.
2. Build a classification model that detects at-risk customers with high recall.
3. Quantify the business value of using the model.
4. Convert findings into actionable retention strategies.

1.3 Stakeholders

Stakeholder	Interest
Retention / marketing team	Receives the list of high-risk customers to contact
Customer-support managers	Understands service issues that cause churn
Finance department	Measures revenue saved versus campaign cost
Senior management	Strategic decisions on pricing, contracts and services

1.4 Business relevance

Churn directly reduces recurring revenue and market share. In this dataset about 26.5% of customers churned. Even a small improvement in retention significantly increases lifetime revenue, which makes churn prediction one of the most widely applied machine learning use cases in telecom, banking and subscription businesses.

1.5 Key Performance Indicators (KPIs)

KPI	Purpose
Churn rate (%)	Baseline business problem measure
Recall (churn class)	Share of real churners that the model catches
Precision (churn class)	Share of contacted customers who are truly at risk
F1-score	Balance between precision and recall
ROC-AUC	Overall ranking quality, independent of threshold
Net revenue saved	Revenue retained minus campaign cost

Recall is prioritised because missing a churner (false negative) loses the customer's annual revenue, whereas contacting a loyal customer by mistake (false positive) costs only a small offer.

2. Dataset and Data Understanding

2.1 Source and relevance

The IBM Telco Customer Churn dataset, publicly available on Kaggle (blastchar/telco-customer-churn) and IBM's GitHub repository. It is a sample dataset created by IBM that models a fictional telecom company's customers. It is relevant because it contains demographics, subscribed services, contract and billing details, and a labelled churn outcome, which is exactly what a churn model requires. It is widely used for learning and benchmarking. Being a sample dataset, results should be re-validated on real operational data before deployment.

- Rows: 7,043 customers
- Columns: 21 (19 features, 1 identifier, 1 target)
- Target variable: Churn (Yes / No)

2.2 Data dictionary

Column	Type	Description
customerID	Identifier	Unique customer ID (dropped before modelling)
gender	Categorical	Male / Female
SeniorCitizen	Binary	1 if customer is a senior citizen
Partner	Categorical	Has a partner (Yes/No)
Dependents	Categorical	Has dependents (Yes/No)
tenure	Numeric	Months the customer has stayed
PhoneService	Categorical	Has phone service

Column	Type	Description
MultipleLines	Categorical	Has multiple phone lines
InternetService	Categorical	DSL / Fiber optic / No
OnlineSecurity	Categorical	Online security add-on
OnlineBackup	Categorical	Online backup add-on
DeviceProtection	Categorical	Device protection add-on
TechSupport	Categorical	Technical support add-on
StreamingTV	Categorical	Streams TV
StreamingMovies	Categorical	Streams movies
Contract	Categorical	Month-to-month / One year / Two year
PaperlessBilling	Categorical	Uses paperless billing
PaymentMethod	Categorical	Electronic check, mailed check, bank transfer (auto), credit card (auto)
MonthlyCharges	Numeric	Current monthly bill
TotalCharges	Numeric	Total amount billed over the tenure
Churn	Target	Yes if the customer left

2.3 Data quality assessment

Check	Finding
Duplicate rows	None
Missing values (null)	None detected directly
Hidden missing values	TotalCharges stored as text; 11 blank entries
Class balance	About 73.5% stayed, 26.5% churned (imbalanced)
Data types	TotalCharges needed conversion from text to numeric

3. Data Preprocessing and Cleaning

Step	Action	Justification
Type conversion	Converted TotalCharges to numeric	Blank strings made the column text type

Step	Action	Justification
Missing values	The 11 blank TotalCharges all belong to customers with tenure = 0 (new customers not yet billed); filled with 0	Logical value, not a guess such as the mean
Duplicates	Checked; none found	Data integrity
Identifier	Dropped customerID	Carries no predictive information and can cause overfitting
Target encoding	Churn mapped to Yes = 1, No = 0	Required for classification
Category cleaning	Merged 'No internet service' and 'No phone service' into 'No'	These are redundant duplicates of 'No'
Outliers	IQR method applied to tenure, MonthlyCharges, TotalCharges; zero outliers found	No removal necessary
Encoding	One-hot encoding (drop first) for categorical variables	Models need numeric input; avoids false ordering
Scaling	StandardScaler on numeric features	Needed for Logistic Regression; keeps features comparable

Encoding and scaling are performed inside a scikit-learn Pipeline and fitted only on training data, which prevents data leakage.

4. Exploratory Data Analysis (EDA)

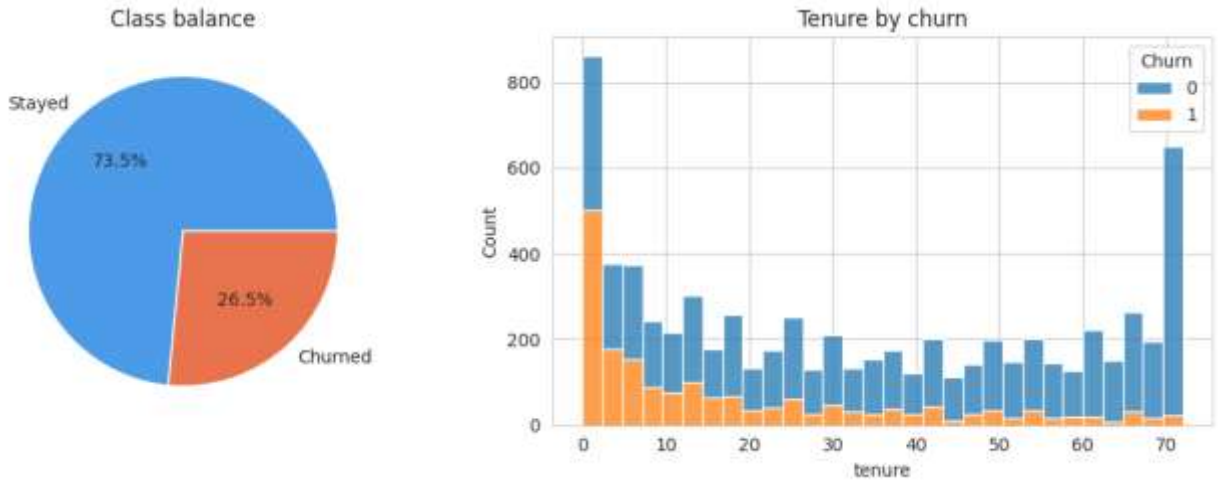


Figure 1 class balance and tenure histogram

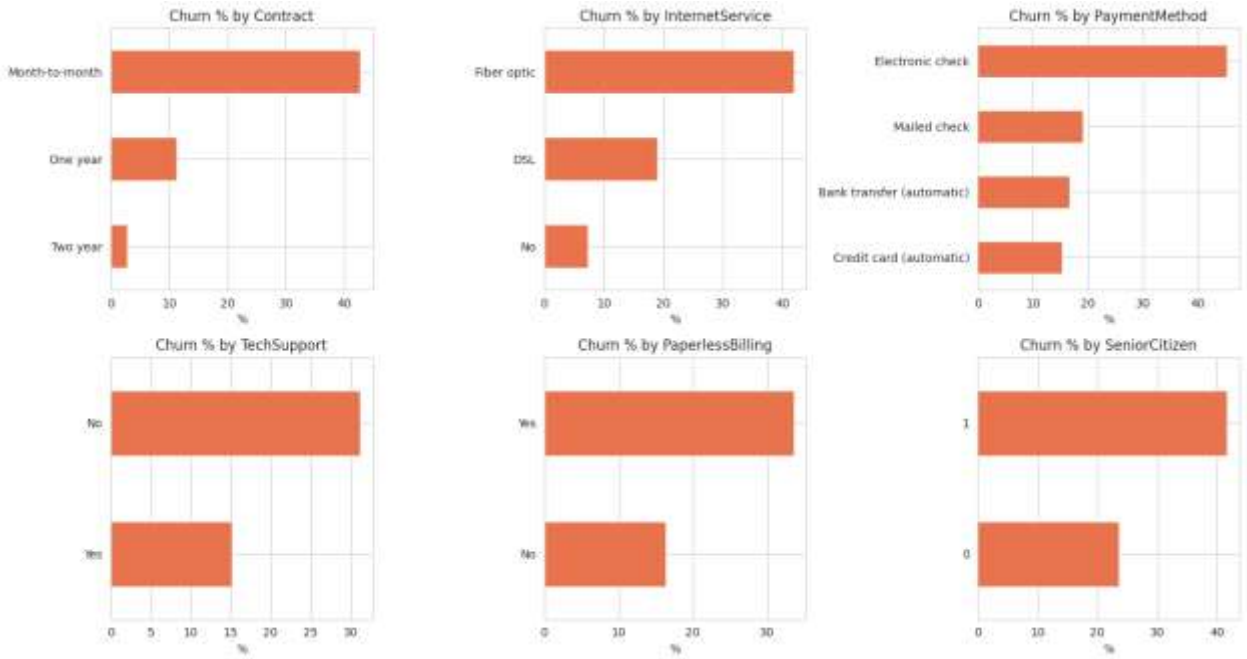


Figure 2: Churn percentage by contract, internet service, payment method, tech support, paperless billing and senior citizen

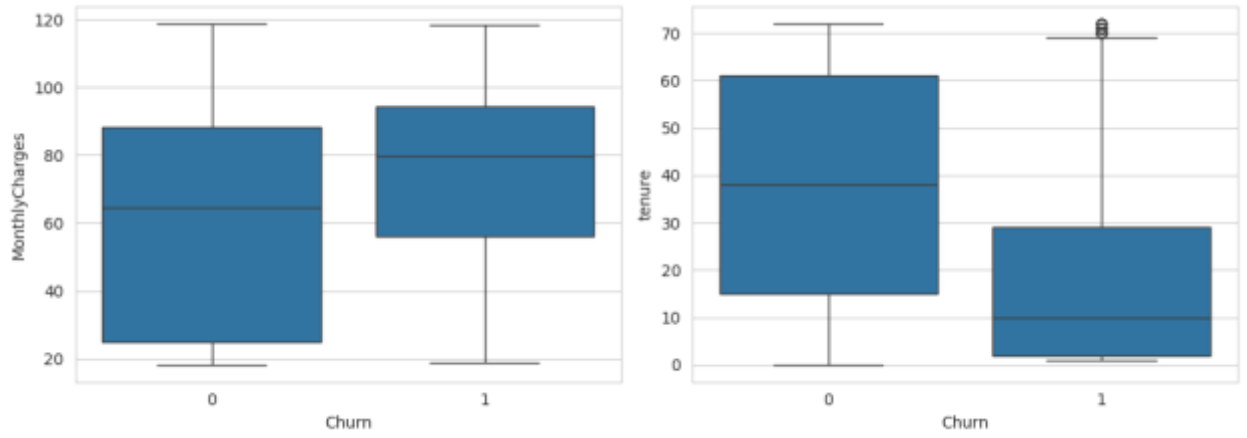


Figure 3: Box plots of monthly charges and tenure for churned vs retained customers

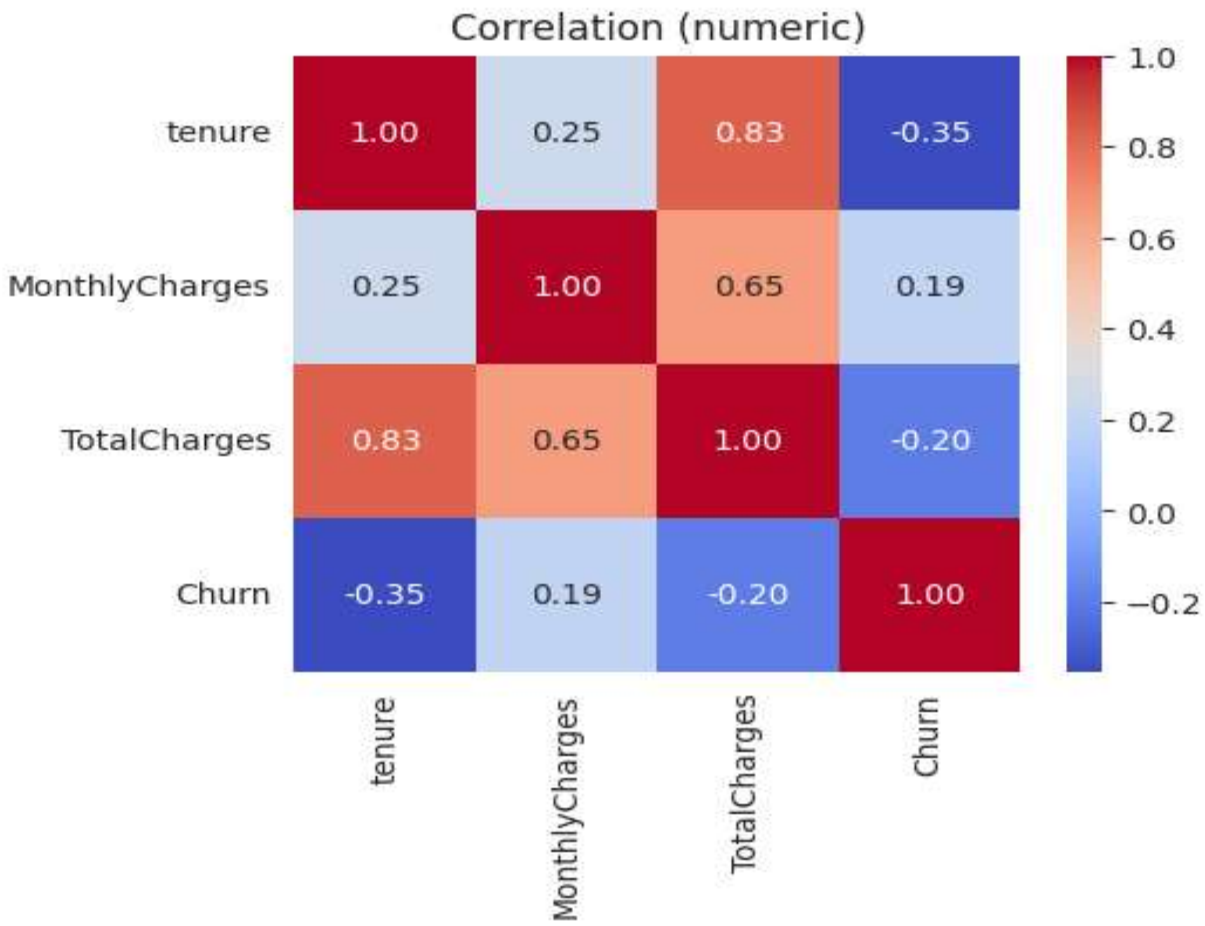


Figure 4: Correlation heatmap of numeric features and churn

4.1 Overall churn

About 26.5% of customers churned, so the dataset is imbalanced. Accuracy alone is therefore misleading: a model predicting that nobody churns would score 73.5% accuracy but be useless.

4.2 Key patterns (churn rate by group)

Feature	Group	Churn rate
Contract	Month-to-month	42.7%
	One year	11.3%
	Two year	2.8%
Internet service	Fiber optic	41.9%
	DSL	19.0%
	No internet	7.4%

Feature	Group	Churn rate
Payment method	Electronic check	45.3%
	Mailed check	19.1%
	Bank transfer (auto)	16.7%
	Credit card (auto)	15.2%
Tenure	0-12 months	47.4%
	13-24 months	28.7%
	25-48 months	20.4%
	49-72 months	9.5%
Tech support (internet users)	No	41.6%
	Yes	15.2%
Senior citizen	Yes	41.7%
	No	23.6%
Paperless billing	Yes	33.6%
	No	16.3%

4.3 Numeric comparison

Group	Average tenure (months)	Average monthly charge
Stayed	37.6	61.3
Churned	18.0	74.4

Churners are newer customers who pay higher monthly bills. TotalCharges is strongly correlated with tenure and MonthlyCharges because it is roughly their product.

4.4 Statistical testing

Chi-square tests were used for categorical features and Mann-Whitney U tests for numeric features against Churn (significance level 0.05). Contract, InternetService, PaymentMethod, TechSupport, OnlineSecurity, tenure, MonthlyCharges and TotalCharges were highly significant. Gender and PhoneService were not significant.

4.5 Insights

1. Month-to-month contracts are the single strongest churn indicator.
2. Most churn occurs in the first year of the relationship.

3. Fiber-optic customers churn about twice as much as DSL customers, suggesting a price or service-quality problem.
4. Customers without technical support or online security churn far more.
5. Electronic-check payers churn most; automatic payment customers are the most loyal.
6. Gender has no meaningful influence on churn.

5. Feature Engineering and Selection

5.1 New features

Feature	Definition	Reason
tenure_group	Tenure binned into 0-1, 1-2, 2-4 and 4-6 years	Churn falls sharply with age of relationship; captures non-linear effect
num_addons	Count of add-on services subscribed	Customers with more services are more attached
avg_monthly_spend	TotalCharges divided by tenure	Historical average bill
charge_gap	MonthlyCharges minus avg_monthly_spend	Indicates recent price increases
autopay	1 if payment method is automatic	EDA showed automatic payers churn far less

5.2 Feature selection

Gender and PhoneService were removed because statistical tests showed no significant relationship with churn, which reduces noise. customerID was dropped as an identifier.

5.3 Data leakage handling

- The train-test split is done before any scaling or encoding.
- Transformations are fitted only on the training set inside a Pipeline, and cross-validation refits them in every fold.
- Threshold selection uses out-of-fold predictions on training data, never the test set.
- The test set is used only once for final reporting.
- No feature uses information from after the churn event; all features are available at prediction time.

6. Model Selection and Implementation

6.1 Methodology

1. Stratified 80/20 train-test split (5,634 training and 1,409 test rows), random_state = 42, preserving the churn ratio.
2. Preprocessing pipeline: StandardScaler for numeric and one-hot encoding for categorical features.

3. Four models trained and compared using stratified 5-fold cross-validation on the training set.
4. Hyperparameter tuning with GridSearchCV on the best candidate.
5. Decision-threshold tuning to maximise churn-class F1.

6.2 Algorithms and justification

Model	Why chosen
Dummy (majority class)	Baseline to prove the models learn something
Logistic Regression	Simple, interpretable, strong benchmark for tabular data
Random Forest	Handles non-linear relationships and interactions; robust to noise
Gradient Boosting	Sequentially corrects errors; usually the best on structured data

Class imbalance was handled with class_weight = 'balanced' (Logistic Regression and Random Forest) and by threshold tuning (final model).

6.3 Hyperparameter tuning

Grid search over n_estimators (100, 200), learning_rate (0.05, 0.1) and max_depth (2, 3) with ROC-AUC scoring. Best parameters: n_estimators = 200, learning_rate = 0.05, max_depth = 2 (cross-validated AUC = 0.848). The shallow trees and low learning rate indicate that a simple, well-regularised model generalises best, reducing overfitting.

7. Model Evaluation and Validation

7.1 Validation strategy

Stratified 5-fold cross-validation on the training data for model comparison; a held-out 20% test set for final unbiased evaluation. Accuracy, precision, recall, F1 and ROC-AUC were all reported.

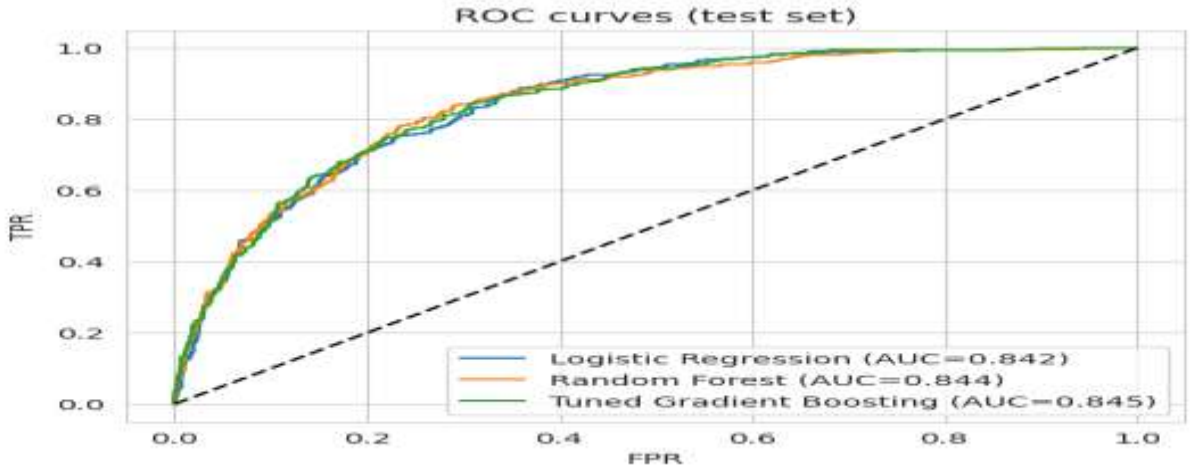


Figure 5: ROC curves of Logistic Regression, Random Forest and tuned Gradient Boosting on the test set

7.2 Cross-validation results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline (majority)	0.735	0.000	0.000	0.000	0.500
Logistic Regression	0.749	0.518	0.799	0.628	0.846
Random Forest	0.781	0.571	0.696	0.627	0.843
Gradient Boosting	0.802	0.669	0.504	0.575	0.846

Gradient Boosting has the highest accuracy but poor recall at the default 0.5 threshold. Accuracy is therefore not a reliable selection criterion here. All three real models have similar AUC (about 0.845), so Gradient Boosting was tuned and its threshold optimised for churn detection.

7.3 Final test-set performance (tuned Gradient Boosting, threshold = 0.32)

Class	Precision	Recall	F1	Support
Stayed	0.89	0.77	0.83	1,035
Churned	0.54	0.75	0.63	374

Overall accuracy: 0.77. ROC-AUC on the test set: about 0.85.

7.4 Confusion matrix

	Predicted: Stayed	Predicted: Churned
Actual: Stayed	801 (true negatives)	234 (false positives)
Actual: Churned	94 (false negatives)	280 (true positives)

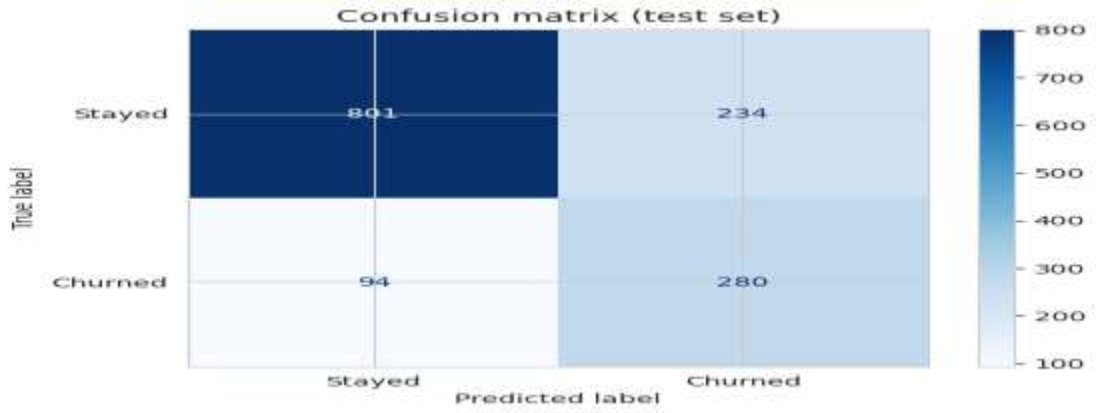


Figure 6: Confusion matrix of the final model on the test set (threshold = 0.32)

7.5 Interpretation

- The model correctly finds 280 of the 374 real churners (75% recall).
- It raises 234 false alarms; these customers receive an unneeded offer, which is a cheap error.
- The threshold was lowered from 0.5 to 0.32 deliberately: the business loses far more from a missed churner than from an extra offer.
- ROC-AUC of about 0.85 means that a randomly chosen churner is ranked as riskier than a randomly chosen loyal customer about 85% of the time, showing strong discrimination.
- Cross-validated and test results are consistent, which suggests the model is not overfitted.

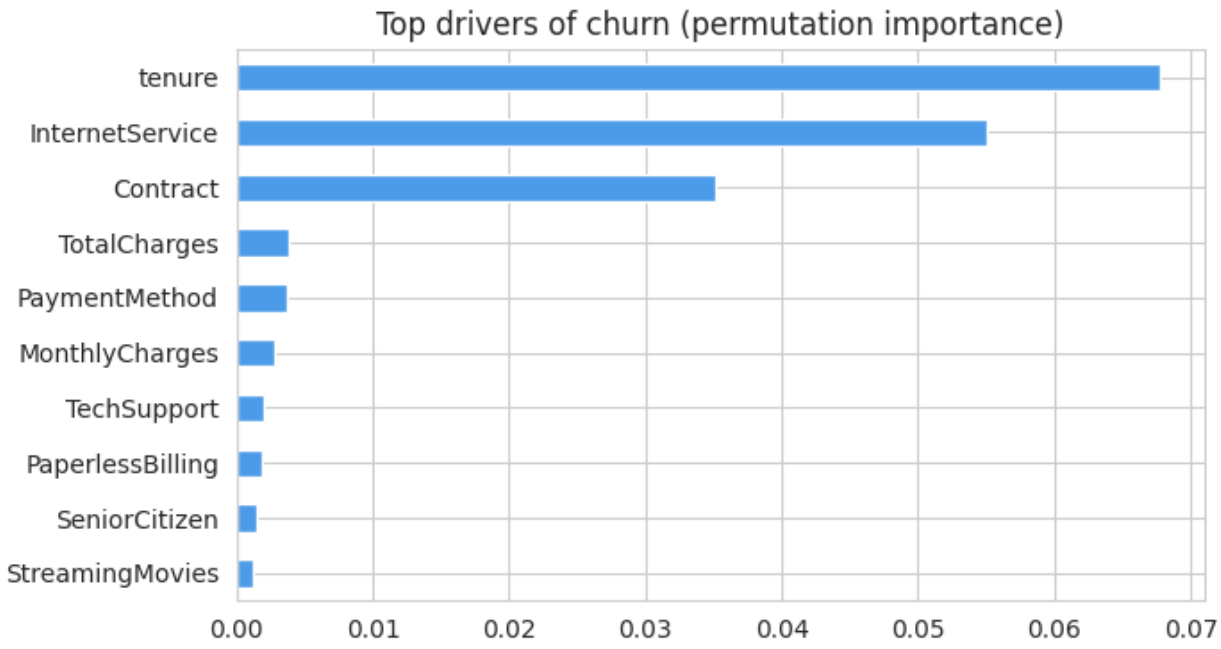


Figure 7: Top 10 churn drivers by permutation importance

7.6 Explainability

Permutation importance (ROC-AUC based) on the test set shows that contract type, tenure, internet service type, monthly charges and add-on services such as tech support and online security are the main drivers, which agrees with the EDA findings and makes the model's behaviour business-interpretable.

8. Business Insights and Recommendations

8.1 Financial impact simulation

Assumptions (adjustable): annual revenue per customer equals average monthly charge multiplied by 12 (about 777); retention offer costs 50 per customer; 30% of contacted churners are retained.

Item	Value
Customers contacted by campaign	514

Item	Value
Real churners reached	280
Revenue saved	65,280
Campaign cost	25,700
Net benefit	39,580
Revenue at risk if no action is taken (374 churners)	290,650

The figures use assumed values and are indicative, not guaranteed.

8.2 Recommendations

No.	Recommendation	Evidence
1	Offer discounts to move month-to-month customers to one- or two-year contracts	Churn 42.7% vs 2.8% for two-year contracts
2	Run onboarding and loyalty programmes in the first 12 months	47.4% churn in the first year
3	Bundle or trial TechSupport and OnlineSecurity for customers without them	41.6% vs 15.2% churn
4	Investigate fiber-optic pricing and service quality	41.9% churn vs 19.0% for DSL
5	Incentivise automatic payments instead of electronic check	45.3% vs about 15-17% churn
6	Score customers monthly and send the top-risk list to the retention team	Model identifies 75% of churners

8.3 Limitations and future work

- The data is a single snapshot with no time dimension, so seasonality and trends cannot be modelled.
- It is a sample dataset; real-company data may behave differently.
- Business assumptions (offer cost, success rate) are estimates.
- No customer-satisfaction, complaint or call-centre data is available.
- Future work: SHAP explanations, XGBoost / LightGBM, SMOTE oversampling, survival analysis for time-to-churn, and a cloud-deployed API with monitoring and periodic retraining.

9. Documentation, Code Organisation and Presentation

9.1 Project structure

File	Purpose

File	Purpose
churn_project.ipynb	Complete notebook, sections aligned to the evaluation criteria
app.py	Streamlit web app for live prediction demo
churn_model.pkl	Saved trained model and threshold
Telco-Customer-Churn.csv	Dataset
requirements.txt	Python dependencies
README.md	Project overview and run instructions

9.2 How to run

1. Install dependencies: pip install -r requirements.txt
2. Open and run all cells of churn_project.ipynb (creates churn_model.pkl).
3. Launch the demo: streamlit run app.py

9.3 Deployment demonstration

The Streamlit app lets a user enter a customer's details and returns the churn probability with a High-risk or Low-risk recommendation. Deployment is shown as a demonstration and is not a separate marking category.

9.4 Individual contributions

Team member	Contribution
Vaishnavi Bhise	problem definition, data understanding, documentation
Srushti Jamdar	preprocessing, EDA and visualisations
Pranjal Dhamane	feature engineering, model building

10. Conclusion

The project delivered a complete data science lifecycle for a real business problem: from problem definition and data understanding to modelling, evaluation, business recommendations and a working demo. The tuned Gradient Boosting model detects 75% of churning customers with an ROC-AUC of about 0.85, and the analysis shows that contract type, early tenure, fiber-optic service, lack of support add-ons and electronic-check payment are the main churn drivers. Acting on these findings with targeted offers can retain revenue at a cost far below the revenue at risk.

References

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